

YANG WANG

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196 Tiber Way ◇ St Charles, MO 63301

EDUCATION

Purdue University, West Lafayette

Aug. 2015 – Jun. 2021

Ph.D. in Electrical Engineering

Thesis: Design and Implementations of Open-Source Ag IoT Devices for Farm Machinery Data Acquisition and Integrated Analytics

Purdue University, West Lafayette

Aug. 2010 – Dec. 2014

B.S. in Electrical Engineering

EXPERIENCE

The Climate Corporation

Nov. 2022 – Present

Senior Data Scientist

St Louis, MO

- Designed and implemented core data models and backend processing components for a Yield Correction API, including development of a Python-based AWS SQS consumer and containerized deployment to AWS ECS with infrastructure provisioned via Terraform.
- Led development of an AWS ECS service to convert shapefiles into an internal geospatial format, increasing conversion success rate from 50% to 95% and eliminating ongoing 3rd-party service costs.
- Architected and implemented a spatiotemporal processing pipeline using connected components algorithms to aggregate vector data, achieving up to 10x data size reduction with minimal fidelity loss.

John Deere Intelligent Solutions Group

Jul. 2021 – Oct. 2022

Senior Navigation Engineer

Torrance, CA

- Refactored and enhanced GNSS receiver (StarFire) firmware in C++, improving reliability and maintainability of embedded navigation systems.
- Implemented state estimation and signal correction algorithms, including multipath mitigation and bias correction techniques.

Purdue University

Aug. 2015 – Jun. 2021

Graduate Research / Teaching Assistant

West Lafayette, IN

- Architected and developed open-source agricultural telematics units (ISOBlue 2) running Yocto Linux for farm machinery data acquisition. Deployed 10+ units across multiple U.S. states and internationally, coordinating directly with farmers and operators.
- Implemented Kalman filtering and spatiotemporal clustering algorithms for GNSS track analysis and harvest productivity modeling.

Spensa Technologies Inc. (Acquired by DTN)

Jan. 2015 – May. 2015

Embedded Systems Engineer

West Lafayette, IN

- Developed embedded firmware and contributed to PCB layout for base station hardware launch.
- Collaborated across hardware and software teams to deliver production-ready embedded systems.

CORE COMPETENCIES

Programming Languages

Python, C, C++, MATLAB, Shell scripting, L^AT_EX, Java, Javascript

Cloud & Containerization

AWS, Docker

Parallel Processing

PySpark

Embedded Systems

Yocto Linux, Android, ISOBUS VT

Network & Messaging Protocols

CAN, J1939, ISOBUS, MQTT, Apache Kafka

Databases

MySQL, TimescaleDB, DuckDB

Observability

Datadog, OpenSearch, Splunk

Version Control Systems

Git, Team Foundation Server

Languages

English and Mandarin (both fluent), French (limited proficiency)